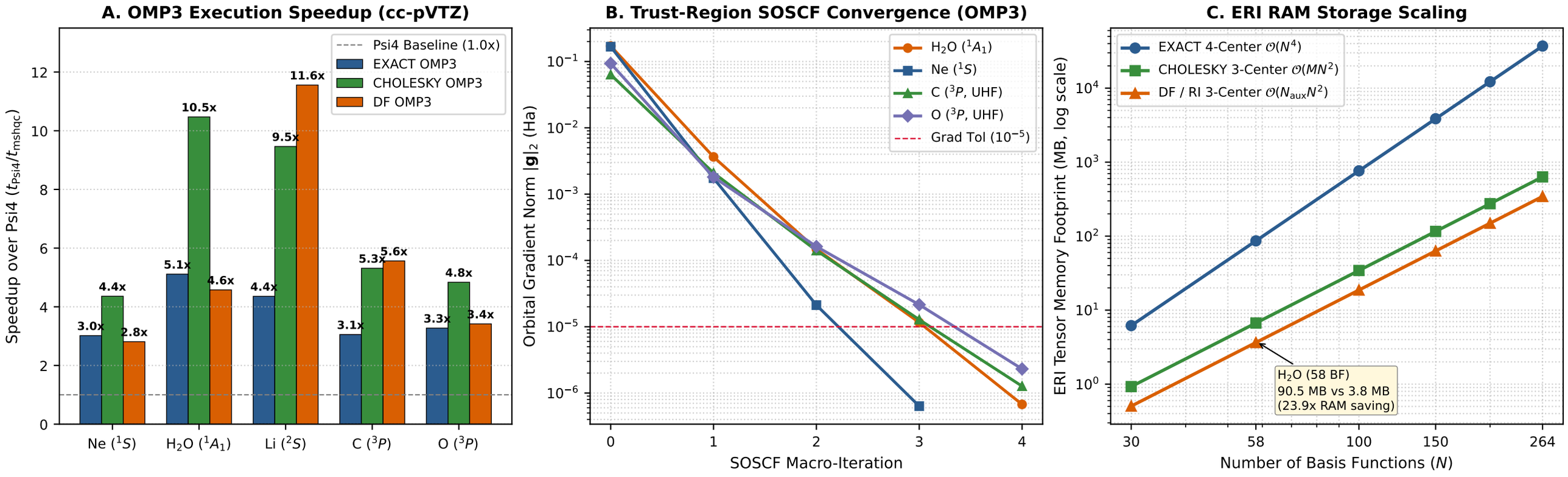
A. OMP3 Execution Speedup (cc-pVTZ)
Speedup over Psi4 (t_{Psi4}/t_{mshqc})
Psi4 Baseline (1.0x)
EXACT OMP3
CHOLESKY OMP3
DF OMP3
3.0x
4.4x
2.8x
5.1x
10.5x
4.6x
4.4x
9.5x
11.6x
3.1x
5.3x
5.6x
3.3x
4.8x
3.4x
Ne (^1S)
H_2O ($^1A_1$)
Li (^2S)
C (^3P)
O (^3P)
B. Trust-Region SOSCF Convergence (OMP3)
Orbital Gradient Norm $|\mathbf{g}|_2$ (Ha)
H_2O ($^1A_1$)
Ne (^1S)
C (^3P, UHF)
O (^3P, UHF)
Grad Tol (10^{-5})
SOSCF Macro-Iteration
C. ERI RAM Storage Scaling
ERI Tensor Memory Footprint (MB, log scale)
EXACT 4-Center $\mathcal{O}(N^4)$
CHOLESKY 3-Center $\mathcal{O}(MN^2)$
DF / RI 3-Center $\mathcal{O}(N_{aux}N^2)$
H_2O (58 BF)
90.5 MB vs 3.8 MB
(23.9x RAM saving)
Number of Basis Functions (N)